Food Chemistry 170 (2015) 336–342

Contents lists available at ScienceDirect

Food Chemistry

journal homepage: www.elsevier.com/locate/foodchem

Isolation and identification of aromatic compounds in Lion's Mane Mushroom and their anticancer activities

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ARTICLE INFO

Article history:
Received 22 May 2014
Received in revised form 7 August 2014
Accepted 14 August 2014
Available online 23 August 2014

Keywords:
Hericium erinaceum
Hericiaceae
Hericerin A
Isohericenone J
Apoptosis
Anticancer activity

ABSTRACT

Lion's Mane Mushroom (*Hericium erinaceum*) is a traditional edible mushroom widely used in culinary applications and as an herbal medicine in East Asian countries. In the present study, two new aromatic compounds, hericerin A (**1**) and isohericenone J (**5**), along with five known compounds, isoericerin (**2**), hericerin (**3**), *N*-De phenylethyl isohericerin (**4**), hericenone J (**6**), and 4-[3′,7′-dimethyl-2′,6′-octadienyl]-2-formyl-3-hydroxy-5-methyoxybenzylalcohol (**7**), were isolated from a methanol extract of the fruiting bodies of *H. erinaceum*. The chemical structures of the compounds were determined from mass spectra and 1D- and 2D NMR spectroscopy. The anticancer effects of the isolated compounds were examined in HL-60 human acute promyelocytic leukaemia cells. Hericerin A (**1**) and hericerin (**3**) significantly reduced cell proliferation with IC_{50} values of 3.06 and 5.47 μM, respectively. These same compounds also induced apoptosis of HL-60 cells, accompanied by time-dependent down-regulation of p-AKT and c-myc levels. These data suggest that compounds **1** and **3** from *H. erinaceum* are suitable for use in potential cancer treatments.

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1. Introduction

Lion's Mane Mushroom (*Hericium erinaceum*), a unique and beautiful white fungus, is a traditional edible mushroom used as an herbal medicine in East Asian countries. The mushroom is called "yamabushitake" and "houtou" in Japan and China, respectively (Kim, Kang, Kim, Nam, & Friedman, 2011). The fruiting bodies of *H. erinaceum* contain various bioactive substances, including aromatic compounds, diterpenoids, sterols, and polysaccharides. Previous pharmacological studies of *H. erinaceum* have identified bioactive components such as polysaccharides that have exhibited immunomodulatory effects (Mizuno, Wasa, Ito, Suzuki, & Ukai, 1992). Hericenones and erinacines are also known to promote nerve growth factor (NGF) synthesis (Ma et al., 2010). Hericenones, which are aromatic compounds, are significantly cytotoxic against A549, SK-OV-3, SK-MEL-2 and HCT-15 cell lines (Kim, Noh, Choi, & Lee, 2012). However, the mechanisms by which such aromatics exhibit anticancer activity are not fully understood. This study, in an effort to identify unique compounds with anticancer properties, resulted in the isolation of two new aromatic compounds, including a new isoindolinone alkaloid, hericerin A, and a new isobenzofuranone derivative, isohericenone J, along with five known compounds, from a methanol extract of the fruiting bodies of *H. erinaceum.* The anticancer activities of the isolated compounds were assessed in HL-60 (leukaemia cells) and HEL-299 (lung fibroblast cells) using an MTT assay. The effects of hericerin A (**1**) and hericerin (**3**) on the induction of apoptosis in HL-60 cell line were also investigated, as was PI3K/AKT activation.

The Bcl-2 family is divided into anti-apoptotic proteins, including Bcl-2 and Bcl-xL, and pro-apoptotic proteins, such as Bax, Bid, and Bak. Bax induces apoptosis by releasing cytochrome *c* from mitochondria. The released cytochrome *c* induces the cleavage of caspase-9 and subsequent cleavage of caspase-3. The activated caspase-3 induces the cleavage of poly(ADP-ribose) polymerase (PARP). In contrast, Bcl-2 inhibits apoptosis, through the suppression of cytochrome *c* release from mitochondria (Hanahan & Weinberg, 2000; Huang, Lai, & Pan, 2005; Zimmermann & Green, 2001).

Phosphatidylinositol 3-kinase (PI3K)/AKT signalling pathways regulate apoptosis, cell survival and cell growth (Yeh et al., 2004). Recent studies have reported that activation of the PI3K/

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http://dx.doi.org/10.1016/j.foodchem.2014.08.078
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AKT pathway promotes the transcription of c-myc, which is an oncoprotein (Kim & Chung, 2002; Yeh et al., 2004). The current study demonstrates the cytotoxicity of two compounds isolated from the mushroom *H. erinaceum* through the induction of apoptosis *via* inactivation of the PI3K/AKT pathway in HL-60 human acute promyelocytic leukaemia cells.

2. Materials and methods

2.1. General experimental procedures

The FT-IR spectra were measured using a JASCO Report-100 infrared spectrometer (JASCO International Co., Ltd, Tokyo, Japan). The NMR spectra were recorded using a JEOL ECA 600 spectrometer (^{1}H, 600 MHz; ^{13}C, 150 MHz; JEOL Ltd, Tokyo, Japan), The LCQ Advantage ion trap mass spectrometer (Thermo Finnigan, San Jose, CA) was equipped with an electrospray ionisation (ESI) source, and high-resolution electrospray ionisation (HR-ESI) mass spectra were obtained using an Agilent 6530 Accurate-Mass Q-TOF LC/MS system. Column chromatography was performed using silica gel (Kieselgel 60, 70–230, and 230–400 mesh; Merck, Darmstadt, Germany) and YMC RP-18 resins, and thin-layer chromatography (TLC) was performed using pre-coated silica-gel 60 F_{254} and RP-18 F_{254S} plates (both 0.25 mm; Merck).

2.2. Fungal material

Dried fruiting bodies of *H. erinaceum* were purchased from an herbal market in Kumsan, Chungnam Province, Korea in August 2013. Its scientific name was identified by one of the authors (Young Ho Kim). A voucher specimen (CNU 13110) was deposited at the Herbarium of the College of Pharmacy, Chungnam National University.

2.3. Extraction and isolation

Dried fruiting bodies (2.5 kg) of *H. erinaceum* were extracted with MeOH (5 L × 3) under reflux. The MeOH extract (320 g) was suspended in water and partitioned with $CHCl_3$, yielding $CHCl_3$ (90.0 g) and water (220 g) fractions. The $CHCl_3$ fraction (90.0 g) was subjected to silica gel (5.0 × 30 cm) column chromatography with a gradient of *n*-hexane–EtOAc–MeOH (25:1:0, 9:1:0, 5:1:0, 2.5:1:0, 1:1:0.1, 1:1:0.3, 0.5:1:0.5; 4 L for each step) to give 8 fractions (Fr. 1A–1H).

The fraction 1C was separated using silica gel (2.0 × 80 cm) column chromatography with a gradient of *n*-hexane–EtOAc (20:1–10:1, 10 L) to give 11 sub-fractions (Fr. 1C-1–1C-11). Fraction 1C-8 was subjected to YMC (1.0 × 80 cm) column chromatography with an MeOH–acetone–H_2O (3:3:1, 4:4:1, 6:6:1; 1.2 L for each step) elution solvent to give compounds **3** (86.2 mg) and **6** (43.6 mg).

Fraction 1D was subjected to silica gel (2.5 × 30 cm) column chromatography with a gradient of *n*-hexane–EtOAc–MeOH (8:1:0.15, 6:1:0.15, 4:1:0.15, 3:1:0.15, 2:1:0.15, 1.5:1:0.15; 2.5 L for each step) to give 10 fractions (Fr. 1D-1–1D-10). Fraction 1D-4 was subjected to YMC (1.0 × 80 cm) column chromatography with an MeOH–H_2O (3.8:1; 1.0 L) elution solvent to give compound **5** (31.2 mg). Fraction 1D-7 was subjected to YMC (1.5 × 80 cm) column chromatography with an MeOH–acetone–H_2O (1.5:1:1, 3:1.5:1, 6:3.5:1, 9:5:1; 1.0 L for each step) elution solvent to give compound **7** (21.4 mg). Fraction 1D-9 was subjected to silica gel (1.0 × 80 cm) column chromatography with *n*-hexane–EtOAc–acetone (3:1:0.15; 750 mL) to give compound **2** (13.4 mg). Fraction 1E was separated using YMC (1.0 × 80 cm) column chromatography with an MeOH–H_2O (1:1, 3:1, 5:1, 7.5:1, 9:1, 15:1; 300 mL for each step) elution solvent to give compounds **1** (7.3 mg) and **4** (11.8 mg).

2.3.1. Hericerin A (*1*)

Yellowish amorphous powder; IR (KBr): ν_{max} 3385, 1687, 1605 cm^{-1}; ^{1}H NMR (CD_3OD, 600 MHz) and ^{13}C NMR data (CD_3OD, 150 MHz), see Table 1; HR-ESI-MS: *m/z* 360.2207 $[M+H]^+$ (calcd. for 360.2169, $C_{21}H_{30}NO_4$).

2.3.2. Isohericeone J (*5*)

Colourless amorphous powder; IR (KBr): ν_{max} 3274, 1747 cm^{-1}; ^{1}H NMR ($CDCl_3$, 600 MHz) and ^{13}C NMR data ($CDCl_3$, 150 MHz), see

Table 1
The ^{1}H and ^{13}C NMR spectroscopic data of compound **1** and **4**–**6**.

	1[a]		**4**[a]		**5**[b]		**6**[b]	
	δ_C[c]	δ_H[d] (*J* in Hz)	δ_C[c]	δ_H[d] (*J* in Hz)	δ_C[c]	δ_H[d] (*J* in Hz)	δ_C[c]	δ_H[d] (*J* in Hz)
1	170.2		172.9		172.2		172.9	
3	49.3	4.34 s	43.3	4.19 s	68.3	5.21 s	70.3	5.23 s
3a	121.4		123.5		127.3		146.0	
4	149.9		150.2		150.3		96.0	6.49 s
5	121.3		121.6		121.7		164.9	
6	159.1		159.2		159.3		117.0	
7	96.3	6.77 s	96.3	6.78 s	98.5	6.93 s	154.5	
7a	130.7		130.7		124.9		104.2	
1′	22.2	3.29 d (7.5)	22.2	3.29 d (7.5)	23.0	3.49 d (7.5)	21.4	3.36 d (7.5)
2′	122.1	5.04 t (7.5)	122.2	5.04 t (7.5)	120.5	5.21 t (7.5)	121.3	5.18 d (7.5)
3′	134.4		134.4		140.2		135.8	
4′	39.6	1.82 t (7.5)	39.6	1.83 t (7.5)	39.7	2.07 m[e]	39.6	1.96 t (7.5)
5′	26.4	1.92 m	26.3	1.92 m	26.3	2.08 m[e]	26.5	2.05 m
6′	124.1	4.93 t (7.5)	124.0	4.92 t (7.5)	123.6	5.00 t (7.5)	124.3	5.06 t (7.5)
7′	130.7		130.4		132.4		131.2	
8′	24.5	1.48 s	24.5	1.47 s	25.7	1.63 s	25.5	1.64 s
9′	14.9	1.65 s	14.9	1.65 s	16.3	1.79 s	15.9	1.77 s
10′	16.4	1.43 s	16.3	1.42 s	17.8	1.56 s	17.5	1.57 s
1″	45.0	3.59 t (7.5)						
2″	59.9	3.69 t (7.5)						
OMe	55.0	3.73 s	55.0	3.73 s	56.2	3.84 s	56.0	3.90 s

Assignments were done by HMQC and HMBC experiments; *J*-values (Hz) are in parentheses.
[a] Measured in CD_3OD.
[b] Measured in $CDCl_3$.
[c] 150 MHz.
[d] 600 MHz.
[e] Overlapped.

Table 1; HR-ESI-MS: *m/z* 339.1598 $[M+Na]^+$ (calcd. for 339.1567, $C_{19}H_{24}O_4Na$).

2.4. Cell culture and reagents

The HL-60 (human acute promyelocytic leukaemia) and HEL-299 (human lung fibroblast) cell lines were obtained from the Korea Cell Line Bank (KCLB) and cultured in RPMI 1640 (Hyclone, Logan, UT) medium supplemented with 10% foetal bovine serum (Hyclone), 100 U/mL penicillin and 100 mg/mL streptomycin (GIBCO Inc., Grand Island, NY) at 37 °C in a humidified 5% CO_2 atmosphere.

2.5. Cell viability assay

The effects of isolated compounds on the proliferation of HL-60 and HEL-299 cells were evaluated using 3-[4,5-dimethylthiazol-2-yl]-2,5-diphenyltetrazolium bromide (MTT) assay (Scudiero et al., 1988). HL-60 (3×10^5 cells/mL) and HEL-299 (1×10^5 cells/mL) cells were seeded on 96-well microplates of 200 μL. After 24 h, the cells were treated with compounds (0.01, 0.1, 1, 10, 50, and 100 μM) for 72 h. At the end of the experimental incubation, the cells were treated with 50 μL (5 mg/mL) MTT dye and incubated at 37 °C for 4 h. The medium was aspirated and replaced with 150 μL/well dimethyl sulfoxide to dissolve the formazan solution. Cell viabilities were determined by measuring the absorbance at 540 nm using a microplate ELISA reader (Amersham Pharmacia Biotech, Piscataway, NJ). Concentration (*x*-axis) – cell viability (%; *y*-axis) curves for compound-treated cancer cells were obtained. We determined the IC_{50} values (compound concentration resulting in a 50% inhibition of growth).

2.6. Flow cytometric analysis of apoptosis

The effects of compounds **1** and **3** on cell cycle phase distribution were analysed by flow cytometry after staining the cells with propidium iodide (PI). HL-60 cells (3×10^5 cells/mL) were treated with compounds **1** and **3** (IC_{50} concentrations) for 24 h and 48 h. The treated cells were washed two times with 0.01 M phosphate buffered saline (PBS; NaCl 0.138 M; KCl 0.0027 M; pH 7.4) and fixed with 70% ethanol for 30 min at 4 °C. The fixed cells were washed twice with cold PBS, incubated with 50 μg/mL RNase A at 37 °C for 30 min, and stained with 50 μg/mL PI solution (Sigma) in the dark for 15 min at 37 °C. The stained cells were analysed using an EPICS-XL FACScan flow cytometer (Beckman Coulter, Miami, FL). The proportions of cells in G0/G1, S, G2/M phases were represented as DNA histograms. Apoptotic cells with hypodiploid DNA content were measured by quantifying the sub-G1 peak in the cell cycle pattern. For each experiment, 10,000 events per sample were analysed, and experiments were repeated 3 times.

2.7. Morphological analysis of apoptosis by Hoechst 33342 staining

For detection of apoptosis, HL-60 cells (3×10^5 cells/mL) were treated with compounds **1** and **3** (IC_{50} concentrations) for 24 h and 48 h. The cells were incubated in Hoechst 33342 (10 μg/mL medium at final) staining solution at 37 °C for 20 min. The stained cells were observed with an inverted fluorescence microscope equipped with an IX-71 Olympus camera and photographed (magnification ×200).

2.8. Western blot analysis

HL-60 cells (3×10^5 cells/mL) were treated with compounds **1** and **3** (IC_{50} concentrations) for 24 h and 48 h. After treatment, the cells were harvested and washed 2 times with cold PBS. The cells were lysed with lysis buffer (50 mM Tris–HCl [pH 7.5], 150 mM NaCl, 2 mM EDTA, 1 mM EGTA, 1 mM $NaVO_3$, 10 mM NaF, 1 mM dithiothreitol, 1 mM phenylmethylsulfonylfluoride, 25 μg/mL aprotinin, 25 μg/mL leupeptin, 1% Nonidet P-40) and kept on ice for 30 min at 4 °C. The lysates were centrifuged at 15,000 rpm and 4 °C for 15 min. The supernatants were stored at −20 °C until use. The protein content was determined by the Bradford assay (Bradford, 1976). The same amounts of lysates were separated on 8–15% SDS–PAGE gels and then transferred onto a polyvinylidene fluoride (PVDF) membrane (Bio-Rad Laboratories, Hercules, CA) by glycine transfer buffer (192 mM glycine, 25 mM Tris–HCl [pH 8.8] and 20% MeOH [*v*/*v*]) at 100 V for 2 h. After blocking with 5% non-fat dried milk, the membrane was incubated with primary antibody against PARP (1:1000), cleaved caspase-3 (1:1000), Bcl-2 (1:1000), Bax (1:1000), c-myc (1:1000), AKT (1:1000), p-AKT (1:1000) and β-actin (1:5000) antibodies and incubated with a secondary HRP antibody (1:5000; Vector Laboratories, Burlingame, VT) at room temperature. The membrane was exposed on X-ray films (AGFA, Belgium), and protein bands were detected using a WEST-ZOL® plus Western Blot Detection System (iNtRON, Gyeonggi-do, Korea).

2.9. Statistical analysis

Results are expressed as means ± standard deviation (SD) from representatives of three independent experiments. The student's *t*-test was used to evaluate the data with the following significance levels: $^{*}p < 0.05$, $^{**}p < 0.01$ and $^{***}p < 0.001$.

3. Results and discussion

3.1. Identification of compounds 1–7

Seven aromatic compounds (**1–7**) were isolated from the $CHCl_3$ fraction of *H. erinaceum* fruiting bodies. Their structures were identified as hericerin A (**1**), isohericerin (**2**), hericerin (**3**; Kobayashi et al., 2012), *N*-De phenylethyl isohericerin (**4**; Miyazawa, Takahashi, Horibe, & Ishikawa, 2012), isohericenone J (**5**), hericeone J (**6**; Ueda et al., 2008), and 4-[3′,7′-dimethyl-2′,6′-octadienyl]-2-formyl-3-hydroxy-5-methyoxybenzylalcohol (**7**; Miyazawa et al., 2012), based on spectroscopic data, chemical evidence, and comparisons with previous reports (Fig. 1). Of these, hericerin A (**1**) and isohericenone J (**5**) were new compounds.

Compound **1** was isolated as a yellowish amorphous powder. The molecular formula was established as $C_{21}H_{29}NO_4$ by a quasi-molecular ion peak $[M+H]^+$ at *m/z* 360.2207 (calcd. 360.2169) in the HR-ESI-MS spectrum. The IR spectrum showed absorption bands characteristic of a hydroxy group (3385 cm^{-1}), a γ-lactam group (1687 cm^{-1}), and an aromatic moiety (1605 cm^{-1}). The ^1H NMR spectrum (Table 1) contained peaks consistent with an aromatic proton at δ_H 6.77 (s, H-7), two trisubstituted olefinic protons at δ_H 4.93 (t, *J* = 7.5 Hz, H-6′) and 5.04 (t, *J* = 7.5 Hz, H-2′), a methoxy group at δ_H 3.73 (s), six methylene groups at δ_H 1.82 (t, *J* = 7.5 Hz, H-4′), 1.92 (m, H-5′), 3.29 (d, *J* = 7.5 Hz, H-1′), 3.59 (t, *J* = 7.5 Hz, H-1″), 3.69 (t, *J* = 7.5 Hz, H-2″), and 4.34 (s, H-3); and three methyl groups at δ_H 1.43 (s, H-10′), 1.48 (s, H-8′), and 1.65 (s, H-9′). Correspondingly, the ^{13}C NMR and DEPT spectra (Table 1) contained peaks corresponding to a methoxy carbon at δ_C 55.0, six signals in the aromatic region of δ_C 96.3–149.9, four olefinic carbon signals at δ_C 122.1 (C-2′), 124.1 (C-6′), 130.7 (C-7′), and 134.4 (C-3′), and six methylene group and three methyl group signals between δ_C 14.9 and 59.9. The above data and comparisons with NMR spectra of isohericerin (**2**) and *N*-De phenylethyl isohericerin (**4**), suggested that compound **1** contains an isoindolinone skeleton. HMBC correlations between H-7 (δ_H 6.77)/C-1 (δ_C 170.2) and H-3

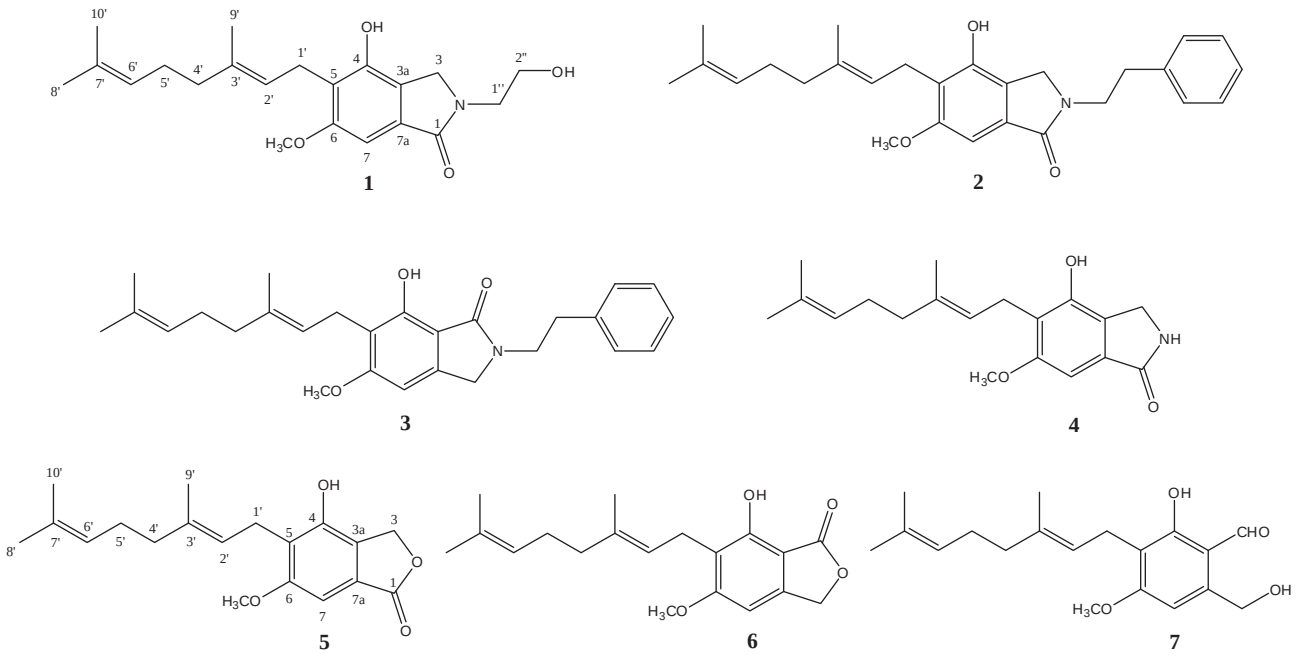

Fig. 1. Structures of compounds **1**–**7** from *H. erinaceum*.

(δ_H 4.34)/C-4 (δ_C 149.9), indicated an isoindoline-1-one substructure (Miyazawa et al., 2012). Correlations between H-1″ (δ_H 3.59)/C-1 (δ_C 170.2), C-3 (δ_C 49.3) indicated that the ethanolic group was located at the nitrogen atom (Fig. 2). Based on these data, the structure of **1** was assigned as shown in Fig. 1 and named hericerin A.

Compound **4** was reported from *H. erinaceum* previously by Miyazawa et al. (2012). However, the ^{13}C NMR data assignments did not match those of *N*-De phenylethyl isohericerin, indicating that the assignments given for C-3a and C-7a should be interchanged. This error is corroborated in another report (Kim et al., 2012). The corrected ^{13}C NMR data of compound **4** are reported (Table 1).

Compound **5** was isolated as a colourless amorphous powder. Its molecular formula, $C_{19}H_{24}O_4$, was determined based on an HR-ESI-MS peak at *m*/*z* 339.1598 $[M+Na]^+$ (calcd 339.1567). The ^{1}H NMR spectrum of **5** showed an aromatic proton at δ_H 6.93 (s, H-7), a methoxy group at δ_H 3.84 (s), and a methylene group at δ_H 5.21 (s, H-3). Other signals correspond to side-chain moieties, which are similar to those of **1**. The ^{13}C NMR spectrum indicated the presence of 19 aromatic carbon atoms, including a carbonyl group (δ_C 172.2), six aromatic carbon atoms (δ_C 98.5–159.3), four olefinic carbon atoms (δ_C 120.5–140.2), a methoxy group (δ_C 56.2), four methylene carbon atoms (δ_C 23.0–68.3), and three methyl carbon atoms (δ_C 16.3–25.7). Both the ^{1}H and ^{13}C NMR spectra of **5** were very similar to those of hericenone J (**6**) (Table 1). The γ-lactone moiety was identified by correlations between H-7 (δ_H 6.93)/C-1 (δ_C 172.2) and H-3 (δ_H 5.21)/C-4 (δ_C 150.3) in the HMBC spectrum (Fig. 2). These results indicate that the chemical structure of **5** differs from that of hericenone J. Thus, compound **5** was named isohericenone J.

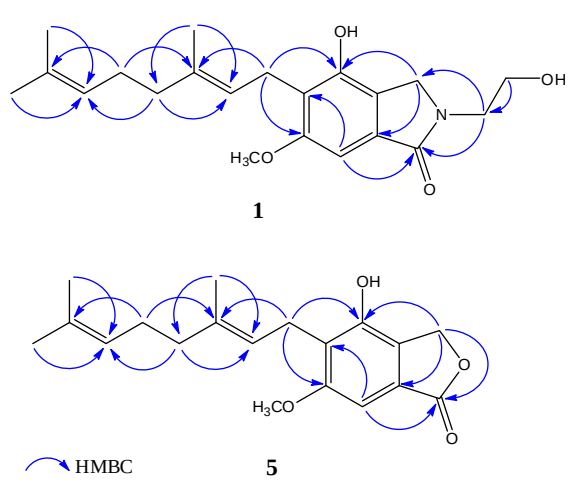

Fig. 2. Key HMBC correlations of compounds **1** and **5**.

3.2. Effects of compounds 1–7 on the growth of HL-60 cells

To evaluate the effects of the isolated compounds on the growth of HL-60 cells, cell viability was assessed using an MTT assay. Compounds **1**, **3**, and **5**–**7** significantly reduced the percentage of viable of HL-60 cells, with IC_{50} values ranging from 3.06 to 5.47 μM. Conversely, compounds **2** and **4** showed weak cytotoxic activities, with IC_{50} values of 59.7 and 62.2 μM, respectively (Table 2). In the development of anticancer drugs, selective cytotoxicity against

Table 2
Inhibitory effects of the isolated compounds on the growth of HL-60 and HEL-299 cells.

Compound	Cell line $IC_{50} \pm SD$[a] (μM)	
	HL-60	HEL-299
1	3.06 ± 0.56	64.61 ± 3.82
2	59.74 ± 0.19	–
3	5.47 ± 0.32	>100
4	62.24 ± 0.70	–
5	4.10 ± 0.21	5.79 ± 0.60
6	4.13 ± 0.20	5.07 ± 0.60
7	4.28 ± 0.19	8.46 ± 0.61
Mitoxantrone[b]	0.075 ± 0.005	–

[a] Values are means ± *SD* (*n* = 3).
[b] Positive control.

cancer cells, and not against normal cells, is crucial. We therefore examined the cytotoxic effects of the isolated compounds in HEL-299 cells, a normal cell line. Compounds **5**–**7** also inhibited the growth of HEL-299, with IC_{50} values ranging from 5.07 to 8.46 μM. However, compounds **1** and **3** inhibited the growth of HEL-299 cells by less than 30%, even at concentrations up to 50 μM. Hence we subsequently investigated cytotoxic mechanisms of compounds **1** and **3** on the apoptosis-induction in HL-60 leukaemia cells.

3.3. Effects of compounds *1* and *3* on the induction of apoptosis

The mechanisms by which compounds **1** and **3** induce cell death were investigated. Apoptotic cell death has typical characteristics, including chromatin condensation, membrane leakage, cell shrinkage and an increased population of sub-G1 hypodiploid cells (Yang, Zhu, Chen, Wang, & Chen, 2007). Flow cytometry was used to measure the induction of apoptosis and cell cycle distribution. Compounds **1** and **3** increased the proportion of the sub-G1 fraction (M1) cells in a time-dependent manner (Fig. 3A). These results show that compounds **1** and **3** induced apoptosis in cells, which is further supported by an increase in the number of apoptotic bodies, readily identifiable by Hoechst 33342 staining in compound-treated cells after 24 and 48 h incubation periods (Fig. 3B). These data indicate that compounds **1** and **3** can induce apoptotic death of HL-60 cells.

3.4. Effects of compounds 1 and 3 on the expression of apoptosis-related proteins and PI3K/AKT signalling

To determine a possible mechanism underlying the induction of apoptosis by compounds **1** and **3**, we examined the expression of Bcl-2 and Bax in HL-60 cells after treatment with **1** and **3**. The expression of Bax increased markedly, whilst the expression of Bcl-2 decreased in a time-dependent manner in the treated cells, with IC_{50} values of 3.06 and 5.47 μM, respectively. We then examined the activation of the caspase cascade in response to compounds **1** and **3**. Expression of the active forms of caspase-3 increased in a time-dependent manner, as demonstrated by the proteolytic cleavage of PARP (116 kDa) to 89-kDa cleavage products (Fig. 4).

The PI3K/AKT signalling pathways regulate apoptosis, cell survival, and cell growth (Kim & Chung, 2002). Activation of AKT contributes to both cell survival and cell growth through

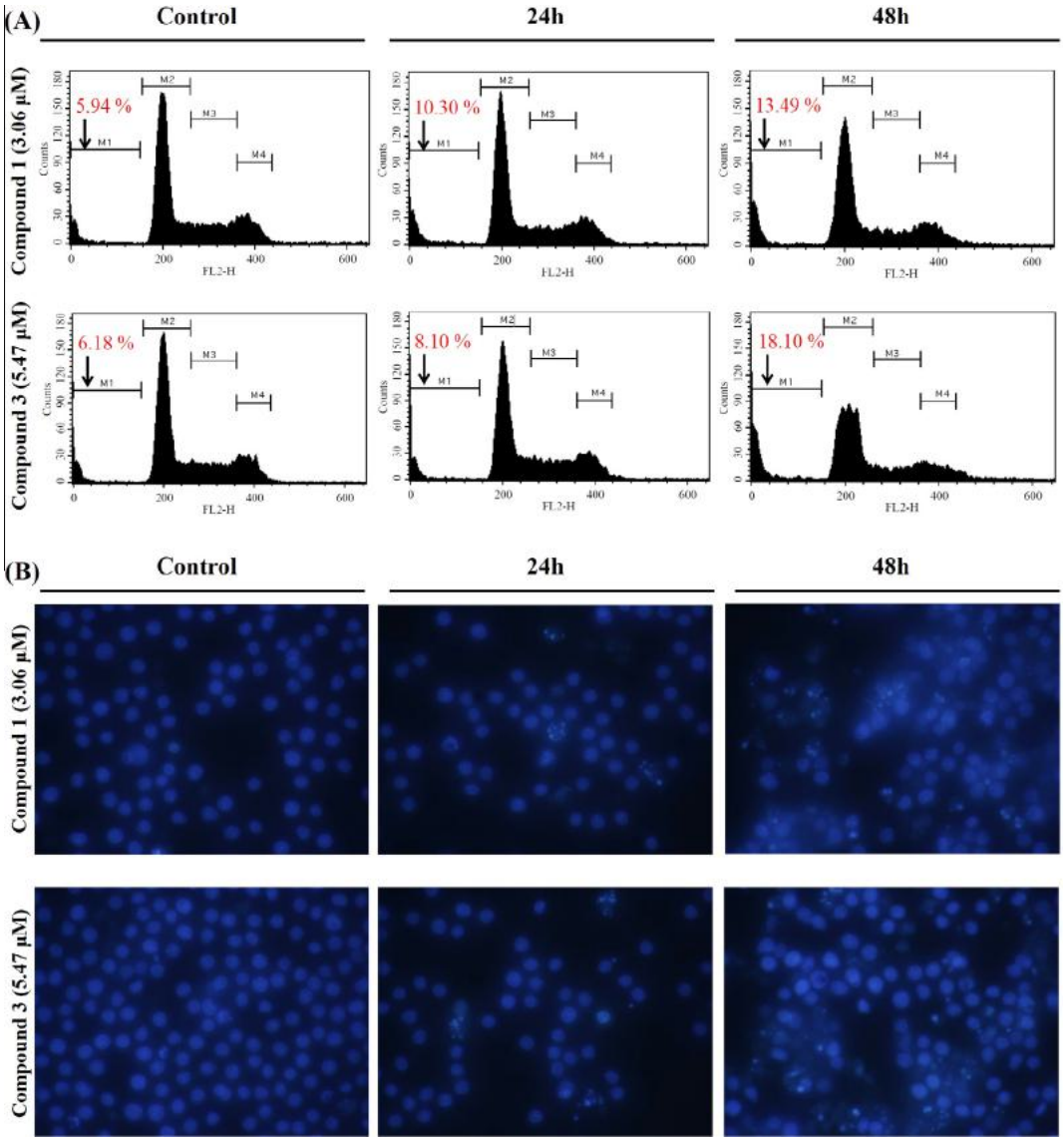

Fig. 3. Compounds **1** and **3** induce apoptosis in HL-60 cells.

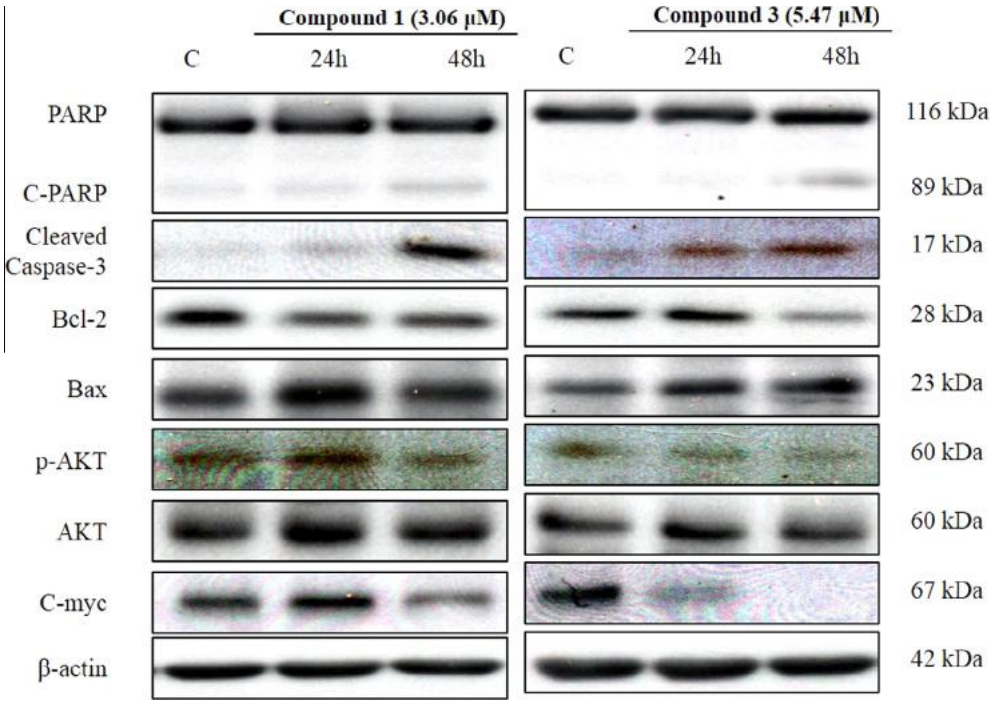

Fig. 4. Effects of compounds **1** and **3** on the levels of apoptosis-related proteins, p-AKT, and c-myc.

c-myc (Hoffman & Liebermann, 2008). The inactivation of the PI3K/AKT by pinosylvin, an anticancer compound, has been reported to attenuate phosphorylation of glycogen synthase kinase 3-β (GSK-3β), which suppresses nuclear translocation of β-catenin, a downstream molecule of PI3K/AKT/GSK-3β signalling. These events by pinosylvin could lead to the down-regulation of β-catenin-mediated transcription of target genes including c-myc (Park et al., 2013). Previous studies have demonstrated that taxol induces apoptosis through the activation of PI3K/AKT pathways (Kim, Juhnn, & Song, 2007; Zhang et al., 2014). The c-myc is overexpressed in 70% of colorectal tumours (Arango et al., 2003). Alterations of chromosome 8, including the c-myc oncogene, have been noted as one of the most common chromosomal abnormalities in prostate cancer progression (Hawksworth et al., 2010). In addition, c-myc is overexpressed in myelogenous leukaemia cells, including HL-60 cells (Dalla- Favera et al., 1983). To investigate such compound-induced intracellular signalling, we used Western blots to examine the phosphorylation of AKT and levels of c-myc. Treatment with compounds **1** and **3** decreased p-AKT levels in conditions that induced apoptosis in the HL-60 cell line. Furthermore, the down-regulation of p-AKT was accompanied by the down-regulation of c-myc (Fig. 4). These findings demonstrate that the apoptosis-inducing effects of compounds **1** and **3** are mediated by the down-regulation of p-AKT and c-myc.

4. Conclusions

In summary, seven aromatic compounds were isolated from the $CHCl_3$ fraction of *H. erinaceum* fruiting bodies. Hericerin A (**1**) and isohericeone J (**5**) were identified as new compounds. Our study demonstrates that aromatic compounds are major active components in the $CHCl_3$ fraction of *H. erinaceum*. Hericerin A (**1**) and hericerin (**3**) inhibited the growth of HL-60 leukaemia cells by the induction of apoptosis, which occurred through the modulation of known apoptosis-related proteins. In addition, the induction of apoptosis was accompanied by the down-regulation of p-AKT and c-myc in a time-dependent manner. These data suggested that aromatic compounds from *H. erinaceum* fruiting bodies have potential as cancer treatments.

Acknowledgements

This study was supported by the Priority Research Center Program (2009-0093815) through the National Research Foundation of Korea (NRF) funded by the Ministry of Education, Science and Technology, Republic of Korea.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.foodchem.2014.08.078.

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